Data centers, environmental health and responsibility

Environmental metrics and U.S. state evidence, with Indiana and California policy applications

86	**50**	**23**
included source records	states in the electricity model	campus water records

Version 1.2 | Evidence cutoff: 1 October 2026

Prepared as a research and policy decision aid. This review documents verified sources and substantial evidence gaps. It is not an exhaustive search of all published literature, a registered systematic review, or a facility-specific health-impact assessment.

Accompanying materials: interactive dashboard, metric framework, California policy register, CivicShield crosswalk, state and campus data, search log, BibTeX and RIS bibliography.

Structured abstract

Background. Data centers connect digital innovation to electricity, water, materials and local infrastructure. Their footprint varies with workloads, cooling technology, electricity supply and the location of affected communities. Environmental responsibility therefore requires more than a favorable power-usage-effectiveness ratio or an annual renewable-energy claim.

Objective. Map published scientific evidence and selected public records; identify measurable indicators for U.S. state comparison; investigate Indiana water claims and local resistance; and define conditions under which additional useful computing could remain within an explicit carbon budget.

Methods. A rapid structured web search and source-following exercise was conducted on 1 October 2026. The included evidence register contains 86 records, including a 23-record core literature corpus of varied publication types. Public-source evidence adds a 50-state electricity model, reported water balances for 23 Google campuses in 14 states, and selected local governance documents. Sources were charted by geography, period, boundary, method and limitations. No exhaustive subscription-database search, independent duplicate screening or meta-analysis was performed.

Results. Electricity use, direct water loss, upstream water, embodied emissions, air pollution and noise need separate indicators. EPRI's 2026 model places U.S. data-center-plus-cryptocurrency electricity at 184.203 TWh in 2024; this is an estimate, not a utility census. Its Indiana midrange value rises from 0.820 TWh in 2024 to 27.595 TWh in the conditional 2030 medium scenario. These are not AI-only figures. Local Indiana evidence shows why annual projections, shared water-system limits, sewer design and construction dewatering cannot be treated as measured campus consumption. [25] [45] [46] [47] [56]

Conclusion. Efficiency can enable more useful computing, but a lower footprint per task does not guarantee lower total emissions. A defensible no-increase claim requires a defined counterfactual, absolute emissions accounting, workload-quality controls, additional clean supply where appropriate, and protections against transferring impacts to water-stressed or highly exposed communities. This deliverable is a scoping-review foundation, not a claim to have identified all published literature. [9] [7]

1. Review questions and methods

The review asks four questions: (1) which direct and indirect environmental-health pathways are supported by the literature; (2) which metrics are comparable across states and which require facility or watershed resolution; (3) what the Indiana evidence establishes about water use and resistance; and (4) how AI deployment and data-intensive innovation could be constrained by absolute carbon, water and exposure budgets. Global scientific studies inform mechanisms; U.S. sources inform the state map; Indiana receives a purposive case study.

Eligibility included scholarly research, reviews and relevant commentaries available by 1 October 2026; authoritative technical inventories and methodological standards; corporate disclosures with identifiable periods and boundaries; and dated government records or original local reporting needed to establish policy events. English-language accessible records were prioritized. General ICT estimates were used only where their boundaries could be distinguished from data centers. Unsupported social-media estimates, generic commercial summaries and duplicate preprint/published versions were not treated as independent evidence.

Searches used the built-in web search to locate publisher pages, DOI records, PubMed records, author manuscripts, institutional repositories, national-laboratory reports, regulator websites and municipal records. The supplied search-log download preserves query strings from the scholarly, state, Indiana and methods searches. A representative executed query was “Mytton 2021 Data centre water consumption npj Clean Water”; state and policy

searches combined named jurisdictions with water, data centers, moratorium, ordinance and official-site terms. These were targeted queries rather than a reproducible exhaustive database export.

Selection was performed by one automated research workflow with parallel thematic extraction and a subsequent targeted consistency audit. This is not independent duplicate human screening. The workflow deduplicated bibliography identities by DOI, title and URL during integration. The log records selected consolidation and exclusion decisions, but transient search snippets and a complete denominator of all screened records were not preserved. Accordingly, no invented PRISMA identification, exclusion or eligibility totals are shown. The included-record count describes the included register, not all records identified. [58]

Data charting retained authors, year, publication type, geography, methods, findings, units, accounting boundaries, status and limitations. Access varied from full reports and manuscripts to abstracts or indexed government text; these limits are recorded. Critical appraisal was descriptive: direct metering or measurement versus engineering estimate or model, temporal relevance, transparency of assumptions, spatial resolution, corporate interest, completeness and support for causality. No formal intervention risk-of-bias scale or pooled certainty grade was assigned to incompatible study designs.

Synthesis is narrative and stratified by outcome and evidence type. Electricity forecasts are not pooled with observations; corporate campus records are not extrapolated to entire states; municipal rules are not promoted into statewide law. “Not established” denotes a retrieval or evidence gap, not proof of zero impact. Statistical meta-analysis would be misleading across different denominators, workloads, grid years, time horizons and lifecycle boundaries.

2. Measurement framework

Indicator	Unit / denominator	Interpretation and reporting rule
Facility electricity	MWh/year and hourly kW or MW	Report measured electricity and peak demand separately; nominal connection capacity is not annual energy.
IT electricity and PUE	kWh IT; facility kWh / IT kWh	A lower PUE does not establish lower total energy; disclose IT load and utilization.
Direct water withdrawal	m^3/year; daily and monthly peaks	Separate potable, groundwater, surface and reclaimed sources, by operational versus construction phase.
Direct consumptive water	m^3/year; L/kWh IT for defined WUE	State accounting boundary and return flows; water withdrawal minus discharge may not resolve local ecosystem effects.
Indirect water	m^3/year associated with purchased power	Track power-source watersheds and allocation assumptions; avoid double counting direct cooling.
Water scarcity	Monthly watershed / aquifer indicators	Screen with hydrological data; verify with local recharge, drought, competing-use and groundwater evidence.
Operational emissions	t CO2e/year and kg CO2e/useful task	Disclose electricity inventory method, factor year, fuels and refrigerants; preserve both totals and intensity.
Lifecycle emissions / e-waste	t CO2e and tonnes retired equipment	Include hardware manufacturing, buildings, lifetime, repair, reuse and traceable treatment.
Air exposure and noise	PM2.5, NOx; ug/m^3; LAeq, Lnight, dB(C)	Measure receptor locations, generator hours and tonal/low-frequency sound; emissions are not diagnosed disease.
Distribution / responsibility	Costs, affected populations, compliance	Publish local infrastructure allocation, complaints, response times, remedies and binding limits.

Metric definitions follow the water and lifecycle literature; specific accountability requirements are recommendations of this synthesis. Aqueduct is useful for prioritization but is not a substitute for a local hydrological assessment. EPA eGRID supports annual electricity inventories; it does not turn an annual state average into an hourly marginal emissions factor. [2] [3] [19] [62] [59]

3. Scientific evidence and critical synthesis

What the literature establishes

Datacenter impacts are measurable, geographically uneven and sensitive to accounting boundaries. Energy efficiency is consequential, but an improved ratio does not establish an absolute reduction. Masanet et al. (2020; L01) modeled 205 TWh of global datacenter electricity in 2018, 6% above 2010 despite 550% growth in compute instances. This is evidence that efficiency and infrastructure changes can decouple service growth from electricity growth over a specified historical interval. It should not be used as a current consumption estimate or proof that subsequent AI expansion will remain decoupled.

Different published totals can also share the same underlying assumptions. Mytton and Ashtine (2022; L11) traced 258 energy estimates from 46 original publications through 676 sources. Their provenance review makes a methodological point for the dashboard: display original sources and dates, distinguish measured electricity from modeled stock estimates, and do not treat repetitions of one estimate as independent corroboration. Counts of proposed facilities and interconnection requests are not operational electricity consumption.

Water accounting requires an additional spatial boundary. Mytton (2021; L02) distinguishes facility water from electricity-related water and explains why site WUE alone is incomplete. Water withdrawal is intake; consumptive use is the portion not returned to the relevant water system. Neither permits nor utility service capacity demonstrate actual consumptive use. A locally meaningful inventory needs withdrawals, returns, sources, seasonal peaks and watershed conditions, with the WUE denominator stated explicitly.

Siddik, Shehabi and Marston (2021; L03) linked a 2018 U.S. datacenter inventory to electricity and utility systems. Approximately one-fifth of the direct water footprint originated in moderately or highly stressed watersheds; nearly half of servers relied partly or wholly on power plants in stressed regions. Their operational model excludes manufacturing and is not a present-day facility census. Nevertheless, it explains why state borders alone cannot contain a facility's environmental footprint: electricity imports move part of the water and carbon burden elsewhere.

Li et al. (2025; L05) estimate 5.4 million liters of operational water for a GPT-3 training example, including 0.7 million liters on site. Those are modeled quantities, not disclosed training-run water meters. The published CACM article supersedes the 2023 preprint for bibliography purposes. Its carbon-water argument also matters operationally: choosing the lowest-carbon hour or region does not necessarily select the lowest-water option. Universal claims that a prompt always consumes a fixed bottle fraction erase model, output-length, grid and cooling differences.

AI, infrastructure and the life cycle

Task-level benchmarks support using the smallest suitable system, while imposing limits on comparisons. Luccioni, Jernite and Strubell (2024; L06) report mean electricity per 1,000 inferences of 0.002 kWh for text classification, 0.047 for text generation and 2.907 for image generation. Their text-generation benchmark emits only 10 new tokens per input. These values are useful illustrations of workload variation, not current universal chatbot coefficients or proof of equivalent task quality. Procurement benchmarks should compare systems on the same task, output requirements, error tolerance and complete service boundary.

Training-only carbon ledgers are incomplete. Luccioni, Viguier and Ligozat (2023; L04) estimate 24.7 tonnes CO2e for BLOOM's final training dynamic electricity, versus 50.5 tonnes when adding the study's broader equipment and operational allocation. Gupta et al. (2021; L19) similarly show why manufacturing and infrastructure deserve attention as operational carbon decreases. Their relative contributions vary with device, service lifetime and electricity supply; neither paper licenses one embodied-carbon percentage for every facility.

Material throughput creates another boundary. Wang et al. (2024; L10) project 1.2-5.0 million tonnes of cumulative generative-AI e-waste during 2020-2030, with 16-86% reductions under circular-economy scenarios. The range is conditional, not an observed annual waste total. Longer useful life, redeployment and documented downstream treatment therefore belong beside operational efficiency in a responsibility assessment. A hardware upgrade requires comparing avoided operational emissions with the new equipment's manufacturing burden.

The research literature offers methods, not guaranteed savings. Henderson et al. (2020; L17) develop instrumented energy/carbon reporting; Strubell et al. (2019; L16) draw attention to development and tuning costs; Schwartz et al. (2020; L08) argue for efficiency alongside accuracy. Dodge et al. (2022; L18) demonstrate operational-carbon variation across cloud regions and times. Together they support measuring full workloads before scheduling, caching, batching, quantizing or replacing models; they do not establish that every optimization preserves quality or

decreases total annual demand.

Environmental health and responsibility

The datacenter-specific health literature remains much weaker on causal outcomes than the footprint literature. Tao and Gao (2025; L13) is a research-agenda letter; Gour, Ortiz and Maibach (2026; L23) is an exploratory Virginia mini review. Both identify credible exposure pathways and substantial empirical gaps. Neither directly measures disease incidence attributable to a particular site. Indoor occupational noise levels must not be presented as outdoor residential exposure measurements.

Han et al. (2026; L14) model pollution-related public-health costs and propose health-informed computing. The June 2026 version reports CACM acceptance and an upper projected burden exceeding $20 billion annually in 2028, with about seven times the national household-average burden in the most affected counties. These are modeled outcomes. Earlier versions used different horizons and inequality comparisons; quoting their claims as current would be misleading.

The most recent health-equity synthesis, Mithu et al. (2026; L22), extends attention to workers, critical-mineral extraction and e-waste communities; only its abstract was assessed here. Ngata et al. (2025; L12) provide a preliminary Northern Virginia community case, while Hogan (2015; L20) examines water and infrastructure governance in Utah. These sources justify evaluating participation, access to records, cost allocation and distribution of burdens alongside technical performance. They cannot establish the current legal status of an Indiana moratorium; that requires dated municipal documents.

A defensible route to additional computing within a carbon budget

An absolute emissions constraint is more demanding than a lower footprint per computation. Kaack et al. (2022; L09) distinguish computing impacts, application effects and wider system changes. Consequently, an AI climate-benefit claim needs a specified counterfactual and measured avoided emissions, minus computing, hardware and induced-demand effects. An efficiency gain that lowers costs and expands use can leave total emissions higher. No reviewed work establishes that arbitrary AI growth can occur with literally no environmental footprint.

Recent reviews underscore implementation gaps. Verdecchia et al. (2023; L07) synthesize 98 Green AI studies, heavily oriented toward training and laboratory work. Chong et al. (2026; L21) review 64 cooling-control studies: only 18 report PUE, three WUE and one CUE. Their reported energy savings cannot be treated as a pooled expected reduction. A local pilot should retain temperature, reliability and water constraints and compare independently metered outcomes against a credible baseline.

Xiao et al. (2025; L15) combine U.S. server-expansion, cooling and grid scenarios. They identify substantial combined mitigation potential but remaining footprints and difficult siting constraints. Their state rankings depend on future grids and hydropower-water allocation; they are scenarios, not permit recommendations.

The following dashboard policy is an analytical recommendation from this synthesis: set annual gross operational and lifecycle carbon budgets, plus site-specific seasonal water caps and exposure limits. Publish both annual totals and service-normalized intensities. Track emissions from actual electricity supply separately from contractual renewable claims; document new clean supply and hourly matching where available. Verify that scheduling does not transfer burdens into stressed watersheds or highly exposed communities. Require generator operating hours and emissions, boundary noise monitoring, public complaints and remedial responses. Reserve low-impact computing capacity for applications with demonstrable public value, and reassess rebound as demand grows. A claim of “no increase” is credible only when the complete measured budget remains within its defined baseline.

4. U.S. state evidence: comparison without false equivalence

The dashboard's main series is EPRI's February 2026 dataset: all 50 states plus a national total, with historical model years through 2024 and conditional projections from 2025 onward. The District of Columbia is not included and is not assigned zero. The model covers small and large data centers and cryptocurrency mining. Its 2024 low-high bounds express source variation, while future scenarios vary development assumptions. They are not statistical confidence intervals. The national total differs by 0.002 TWh from the sum of rounded state rows. [25] [26]

The 2024 midrange estimate is highest in Virginia (33.947 TWh) and Texas (33.539 TWh); Oregon is 11.005 TWh and Indiana 0.820 TWh. These quantities measure modeled electricity volume, not environmental harm or value created. A smaller load in a water-constrained location or near vulnerable residents can create greater local concern than a larger load elsewhere. Source-reported 0.000 TWh values must not be interpreted as proven absence of facilities. [25] [62]

National estimates provide context but are not interchangeable. IEA's April 2026 global report estimates 485 TWh for 2025 and projects 950 TWh in 2030, around 3% of world electricity. LBNL's "2025 Update," released in June 2026, projects 11.8% of U.S. electricity in its 2030 reference case, with 9.5-15.3% across its reported scenarios. They differ in geography, denominator, data vintage, coverage and model assumptions. The older LBNL 2024 report remains useful for its explicit 2023 water estimates; it should not be presented as the latest electricity outlook. [57] [40] [39]

Virginia - Water and air monitoring

2023 utility records estimate 2.1 billion gallons of water use; more than one-third reclaimed. Most buildings use modest volumes but 11 exceeded 50 million gallons. JLARC recommended water estimates and sound modeling in siting reviews; regional air monitors cannot resolve local generator exposure. **Interpretive limit:** Water figure is water use, not a pure net consumptive total; incomplete but large-majority facility coverage. Recommendations are not enacted law. Evidence date: 2024-12-09. [25] [27]

Oregon - Water transparency and ratepayer responsibility

The Dalles has conflicting corporate withdrawal and municipal supply quantities; reconcile the accounting boundary before combining records. POWER Act created a separate class; PGE Schedule 96 implementation assigns service costs and clean-supply obligations to large loads. Sept 2026 government report says PGE residential rates fell 1.3% while data-center rates rose 29%. **Interpretive limit:** Facility water is not Oregon total. Rate change applies to PGE; does not establish all prices or environmental impacts were neutralized. Evidence date: 2026-09-02. [25] [28] [41] [33] [34] [35]

Arizona - Water-constrained permitting

A campus with very low water volume is not automatically an efficient mature facility; compare cooling design, useful work and operating stage. Phoenix July 2025 Special Permit process addresses noise, grid and safety. Its 2024 large-user ordinance requires conservation plans above 250,000 gal/day and 30% recycled/conserved consumption above 500,000 gal/day. **Interpretive limit:** Municipal thresholds, not Arizona-wide limits. Low direct cooling water can shift burdens to electricity supply; indirect water absent from site figures. Evidence date: 2025-07-02. [25] [28] [29] [30]

Georgia - Reclaimed cooling and local pause

Reclaimed-source cooling illustrates why potable demand and total consumptive losses need separate indicators. Campus measurements appear in the water ledger. Troup County adopted a 90-day moratorium Sept 16, 2025 and extensions; latest verified notice extends through June 14, 2026. Status after that date not established. **Interpretive limit:** Local historical moratorium, not a current statewide ban. Reclaimed water reduces potable demand but still evaporates and may affect downstream flows. Evidence date: 2026-02-17. [25] [28] [31]

Texas - Grid pipeline versus realized demand

ERCOT March 26, 2026 listed 355,830 MW data-center requests within 410,618 MW large-load pipeline. Review recommendation: Interconnection and load forecasting need project readiness screening, flexible-load contracts and deliverable clean supply before energization. **Interpretive limit:** Pipeline capacity is speculative, not metered demand; cannot convert to annual consumption by 8,760 hours/year without utilization/completion assumptions. ERCOT excludes some Texas regions. Evidence date: 2026-03-26. [25] [32]

Ohio - Cost allocation and campus water

Inspect campus water balances alongside electricity service-territory conditions; neither alone establishes full environmental performance. AEP Ohio tariff effective July 23, 2025 uses minimum-demand billing, collateral and exit provisions, with a ramp period up to four years followed by eight years. **Interpretive limit:** Minimum billing is not a clean-power guarantee; exact sliding-scale demand formula matters, and service-territory tariff is not statewide. Evidence date: 2026-10-01. [25] [28] [36]

Washington - Tribal resources and disclosure gaps

State workgroup records tribal concerns about salmon, water supplies and electricity growth; 2026 JLARC found eligible urban centers lack actual electricity reporting. Workgroup recommendations and audit calls for public reporting motivate disclosure conditions; enacted status of individual 2026 bills not assessed. **Interpretive limit:** Do not relabel Northwest region data-center-plus-chip forecasts as Washington-only data-center totals. Evidence date: 2026-10-01. [25] [37] [38]

Iowa - Large evaporative cooling load

Large evaporative cooling loads require basin-seasonal water budgets and drought-day capacity assessment beyond statewide annual averages. Review recommendation: Prioritize basin-seasonal water budgets and drought operating rules in siting and procurement. No moratorium status established in this review. **Interpretive limit:** One campus; large volume may reflect size and utilization. Without IT electricity and watershed stress data it is not an efficiency ranking. Evidence date: 2024-12-31. [25] [28]

Nevada - Scarce-water cooling choices

Cooling choices require campus- and basin-specific assessment. Within-state variation cannot establish technology or efficiency without operational context. Review recommendation: Require facility and basin-specific water caps, reclaimed supply assessment and drought response. No comprehensive current moratorium search completed. **Interpretive limit:** Within-state contrast shows cooling/operation heterogeneity. Volume alone cannot identify technology or maturity. Evidence date: 2024-12-31. [25] [28]

Nebraska - Water burden concealed by state averages

State averages can conceal the effects of clustered campuses. Public monthly water balances and source-water information are priorities. Review recommendation: Priority is public monthly withdrawal/consumption and source-water reporting plus audited resource obligations. No specific moratorium status established. **Interpretive limit:** Two named locations do not describe all Nebraska facilities; basin and seasonal context required. Evidence date: 2024-12-31. [25] [28]

The water ledger deliberately uses a harmonized 2024 campus sample from Google's 2025 report, despite a newer company report being available. Twenty-three named locations in fourteen states have reported withdrawal, discharge and consumption. The methods allow engineering estimates when meter data are unavailable; the limited-assurance scope should not be mistaken for a complete facility census. Differences in size, utilization, maturation, weather and source water make raw volumes unsuitable for ranking state efficiency. [28]

5. Indiana: water, public trust and local governance

The most consequential distinction: water quantities have different boundaries

The Indiana dashboard should present each water datum with a metric definition, facility or system boundary, time period, phase, and evidence status. Withdrawal is water entering a facility from a source; consumption is the portion not returned to that source over the accounting period. Sewer discharge is an outgoing flow. Storage volume and pumping capability are capacities, not use. Temporary construction dewatering requires its own category. Neither design maxima nor daily peaks can be multiplied by 365 and called observed annual consumption.

Indiana's Department of Natural Resources provides a useful starting point: significant withdrawal facilities must register when their aggregate withdrawal capability exceeds 100,000 gallons per day and report annual pumping. Its retrieved documentation lists pump registration information as of August 1, 2025 and recent annual withdrawal files. These records do not automatically isolate the data-center share of a municipal supplier's pumping, and missing entries do not establish zero use. No defensible statewide data-center consumption total was established here. DNR administrative-data documentation

New Carlisle: distinguish the campus, municipal plants and aquifer

In a March 17, 2026 PBS discussion, St. Joseph County economic-development director Bill Schalliol described a 24-million-gallon-per-day limit for two town-managed water plants serving multiple users. That number is not an Amazon meter reading, and it excludes some other aquifer withdrawals such as agricultural irrigation. Accordingly, the dashboard should label it **shared water-system limit**, never **Amazon consumption**. Aquifer recharge estimates also differ among public statements; calculating an exact sustainable surplus without reviewing the original hydrologic model would create false precision. PBS firsthand panel transcript

A July 9 corporate statement published by the regional chamber projects about **18 million gallons annually at full buildout**, with outside-air cooling for 98% of the year. Its plain-language term "water use" does not explicitly distinguish withdrawal from consumptive loss. This is a projected campus annual quantity, not observed operations. Its annualized arithmetic average is about 49,315 gallons/day, but that derived number is not a peak-day requirement. Water-replenishment commitments are separate claims and should not be subtracted from actual local withdrawals to erase physical demand. Amazon statement

IDEM's April 1, 2026 approved IEC-3 South sewer permit provides **55,000 gallons/day future average design flow** and **1,612,000 gallons/day future peak design flow**, a 29.3 peaking factor. Permit page 2 identifies a specific sanitary and industrial wastewater project, with treatment by South Bend. These are sewer-design

quantities, not observed campus withdrawal or consumptive cooling demand. IDEM Permit 26205, p. 2

Corrected August 13 reporting separately describes **31 MGD temporary construction-dewatering approval** and dry-well hardship. DNR attributed failures to drought; a university hydrologist questioned that conclusion. Treat hardship as documented and causation as unresolved. The underlying dewatering approval remains unreviewed. Corrected investigation

For accountability, the useful next measurements are monthly purchased water, daily maxima during heat and drought, metered sewer returns, construction pumping by well, groundwater levels in shallow residential and deeper industrial wells, and the disposition of complaints. These are recommended monitoring requirements, not data already obtained.

Fort Wayne and La Porte: infrastructure context versus operating performance

Fort Wayne's official FAQ places average municipal use at **37 MGD** and treatment capacity at **72 MGD**, with supply from the St. Joseph River. Both numbers concern the city system, not Google. The 35 MGD arithmetic difference is treatment headroom under the stated conditions; it is not proof of ecological or drought-safe headroom. The FAQ also describes workload reduction during grid stress rather than substitution of diesel generators, a locally relevant responsible-AI operating commitment that should be checked against actual dispatch and generator-runtime records. Fort Wayne FAQ

City Utilities separately describes a Google-funded **2-million-gallon storage tank** and transmission infrastructure planned for 2026. The tank's volume belongs under public infrastructure; displaying it as daily data-center consumption would be incorrect. Developer funding is a useful responsibility measure, but long-term maintenance costs and actual capacity benefits need separate evaluation. City Utilities project

Microsoft's June 2026 La Porte announcement commits to closed-loop liquid cooling. This is a design choice, not a demonstrated zero-water or zero-carbon outcome; electricity generation, equipment production, initial fill, maintenance and other site uses still require accounting. A March municipal announcement reports full property-tax payment and allocation of 15% of revenue to schools for 20 years. Fiscal distribution and environmental performance should appear as separate dashboard dimensions, since tax revenue does not measure avoided pollution. Microsoft announcement; La Porte agreement announcement

Local resistance is a dated governance outcome, not a statewide ban

Merrillville's official page establishes a municipal pause on new approvals from **June 1, 2026 through May 31, 2027**. It identifies utility needs, traffic, noise and neighborhood compatibility as questions for study. Boone County's signed **Ordinance 2026-08**, adopted June 15, establishes a pause for new applications in its **unincorporated areas, June 16, 2026 through June 15, 2027**. All three scanned ordinance pages were visually verified; the signed instrument supersedes the initially inconsistent announcement URL. Neither action is an Indiana-wide prohibition. Merrillville; Boone County signed ordinance, pp. 1-3

Miami County illustrates why historical moratoria should not be counted as automatically active. The recorded **May 4, 2026** moratorium ends when data-center zoning rules are adopted. September public-radio reporting links county minutes and reports unanimous adoption on **July 20**, alongside continuing citizen concern about transparency. Classify the May pause as historical/replaced, subject to verification of the final ordinance. Recorded moratorium; Indiana Public Radio

WNDU's September 23 report describes a **6-3 preliminary resolution** starting the process toward a possible future hyperscale ban in unincorporated St. Joseph County. It does not establish an enacted ban. Morgan County reporting documents water concerns around Google's Monrovia campus, but the reported prospective 8 MGD estimate lacks a verified operating boundary here. Keep observed consumption blank and do not infer a Morgan

County, Indiana moratorium from similarly named jurisdictions elsewhere. WNDU; Morgan County Correspondent

A responsible Indiana strategy would condition expansion on transparent monthly accounting, enforceable peak and drought limits, verified low-emission workload flexibility, funded groundwater and noise monitoring, and public complaint resolution. These are recommendations derived from the identified evidence gaps. None of the reviewed Indiana records demonstrates that expanding AI causes no additional lifecycle carbon emissions, and no local exposure-to-disease effect size was established.

6. Implementation: AI progress within an absolute budget

The relevant goal is more useful output while holding a defined environmental budget constant. Literal zero-footprint computing is not established by the reviewed literature. A defensible project can instead test whether incremental annual and lifecycle emissions stay below an agreed baseline, with separate water and health constraints. This requires the same service quality, time horizon and system boundary before and after implementation. Any claimed societal emissions savings need a documented counterfactual and should be reported separately from the operator's gross footprint. [9] [15]

Action	Evidence to collect	Acceptance condition / tradeoff
Avoid unnecessary computation	Usage, quality, retention and baseline IT kWh	Delete redundant jobs and duplicate storage; preserve required research outputs and retention obligations.
Use the smallest adequate model	Task-level accuracy, error severity, latency, joules and output length	Benchmark retrieval, rules, conventional statistics and small models before larger generation. Quality equivalence must be demonstrated.
Batch, cache, quantize and distill	Full service energy, cache hits, model quality, total workload	Savings count only after metering; reduce repeated inference and tuning without hiding extra demand.
Shift flexible work in time or place	Marginal grid signal, water stress, latency and transfer energy	Schedule within service constraints; prohibit shifting into worse water or exposure conditions.
Optimize cooling with safeguards	IT energy, facility energy, intake, evaporation, temperature and reliability	Pilot control systems against an explicit baseline; dry, evaporative and liquid options require site-specific evaluation.
Procure additional clean supply	Hourly matching, deliverability, commissioning and electricity inventories	Annual certificates alone do not demonstrate physical zero-emission operation; retain residual gross emissions.
Extend equipment life responsibly	Embodied carbon, repair, performance and retirement pathway	Replace only when validated lifetime savings outweigh new manufacturing impacts; prevent export of unmanaged e-waste.
Enforce budget and community conditions	Annual carbon totals, seasonal water, receptor noise, backup runtime	Stage expansion against verified capacity; pause or correct operation when binding thresholds are exceeded.

These actions combine literature-supported mechanisms with recommended governance rules, not guaranteed reduction percentages. Cooling-control literature is especially uneven in joint reporting of energy, water and carbon. Public acceptance also cannot be inferred solely from investment or job announcements: a German survey

experiment shows the importance of electricity prices, water strain, energy source and operator nationality, but its numerical effects are not transferable to Indiana. [6] [7] [17] [18] [19] [21] [63]

Worked illustrative example. Begin with 10 GWh/year of IT electricity, PUE 1.30, electricity emissions intensity 0.40 kg CO2e/kWh and direct WUE 0.50 L/kWh IT. This gives 13 GWh facility electricity, 5,200 t CO2e and 5 million liters of direct water. Double useful work while reducing IT energy per unit of work by 40%; set PUE to 1.15, the electricity factor to 0.10 kg/kWh and WUE to 0.20 L/kWh IT. The scenario requires 12 GWh IT, 13.8 GWh facility electricity, 1,380 t electricity CO2e and 2.4 million liters of direct water. Add an illustrative 500 t/year of incremental embodied emissions and the comparison is 1,880 versus 5,200 t. These are user-editable assumptions, not observed savings or a lifecycle neutrality certificate.

The example shows that facility electricity can rise even when the included emissions budget falls. If the new electricity factor instead remains 0.40 kg/kWh, the same scenario plus 500 t embodied emissions reaches 6,020 t/year, exceeding the baseline. If cooling shifts water to upstream power plants, a direct-WUE improvement can also conceal a larger indirect water burden. The lab therefore labels excluded boundaries rather than presenting its result as a complete environmental score.

For electricity-only accounting, the maximum workload multiplier without increasing emissions is (PUE0 × grid factor0) / [(1 - energy saving) × PUE1 × grid factor1]. If additional embodied emissions are included, subtract their annual amount from the baseline emissions budget before dividing by the new emissions per baseline workload. This is a transparent accounting identity; it is not a forecast of achievable efficiency or demand. The interface also reports a separate direct-water comparison.

A practical first-year program starts with facility and task metering, a workload-quality benchmark and public baseline; tests a small set of changes on representative tasks; and then expands only where quarterly absolute budgets and local safeguards hold. Data advances can include open environmental reporting schemas, versioned datasets, reproducible benchmark tasks, transparent provenance, compact specialist models and reuse of already-computed scientific results. Record rebound explicitly: new products, longer outputs, more automated agents and wider user adoption may consume the initial savings.

7. Limitations and research priorities

This rapid scoping synthesis is not an exhaustive systematic review of all published literature. Retrieval was English-focused and relied on web discoverability and accessible records; some full texts were unavailable. There was no preregistered protocol, librarian-reviewed indexed search, complete screening ledger, independent duplicate appraisal or author contact. Selected sources overlap and some technical totals share underlying models. The included corpus therefore supports a traceable research foundation, not a claim of saturation or a publication-ready PRISMA flow diagram.

The main state layer is modeled and includes cryptocurrency mining. Water records are a single-operator campus sample, intentionally harmonized to 2024, with possible engineering estimation. Indiana has no verified comprehensive industry consumption total in this extraction. Local policy searches were selective and dates may be superseded. The Boone County announcement initially had a retrieval anomaly; the signed ordinance was subsequently recovered and verified. Health burden estimates are modeled or synthesized, while direct site-specific epidemiology remains limited. Lack of quantified disease burden is not evidence of no health risk. [13] [14] [22] [23] [50]

To complete a formal systematic review, preregister a protocol; commission a librarian-assisted search of Scopus, Web of Science, PubMed, IEEE Xplore and ACM Digital Library, with appropriate engineering databases and citation chasing; export all records; deduplicate deterministically; use two independent screeners and a dispute

process; log full-text exclusion reasons; and use design-appropriate critical appraisal. Extend beyond English where feasible, rerun the search close to publication, and separate endpoint-specific reviews if a meta-analysis becomes possible. A proposed search block is: ("data cent*" OR datacent* OR "AI server*" OR "artificial intelligence") AND (water OR electricity OR carbon OR "life cycle" OR pollution OR noise OR health OR "environmental justice"). This block is a recommendation for future indexed searching, not a claim that it was run across those databases. [58]

8. Conclusions

Environmental responsibility is demonstrated by transparent boundaries, measured absolute outcomes, credible counterfactuals and enforceable local protections. A low PUE, an annual renewable purchase or a replenishment promise is insufficient on its own. Indiana’s immediate need is clearer facility-level water accounting and trustworthy governance; the national need is to couple additional computing with real capacity, clean electricity, lifecycle stewardship and community-level exposure evidence. The most supportable ambition is useful AI growth within explicit, verified budgets - not an unqualified promise of no footprint.

9. From public questions to a U.S. research agenda

The companion dashboard now provides a plain-language starting point and an editorial selection of ten policy-relevant readings: eight from 2025-26 and two foundations. Selection considers a new measurement approach, a neglected exposure or distributional question, relevance to a U.S. decision, and transparent limitations. This is an annotated discovery shortlist, not a ranking by citation impact or a claim that these are the only important studies. Two added preprints extend the original register; their peer review and final publication have not been verified.

For residents, the starting questions concern bills, water reliability, noise, health and enforceable benefits. Useful evidence is tied to a specific facility, reporting period and affected population. Community oversight should distinguish annual use from peak capacity, promised benefits from realized outcomes, and a permit limit from measured consumption. The dashboard provides a claim decoder and questions to take to a public hearing; these questions are recommendations of this synthesis, not universally applicable legal requirements.

U.S. research priority	Recommended study design	Decision-relevant outcome
Coincident water and power peaks	Link hourly facility intake, IT load and weather to utility capacity and watershed conditions.	Dry-season reliability and who funds treatment or network expansion.
Actual changes in grid emissions	Validate modeled dispatch responses to flexible computing against observed generation and emissions.	Additional emissions avoided per shifted MWh, with water and reliability constraints.
Health and distribution	Combine pre-opening baselines, receptor monitoring and matched comparison communities.	Changes in exposure, uncertainty and distribution across host and power-supply communities.
Tariffs, permits and public benefits	Compare jurisdictions before and after policy changes; account for demand and investment differences.	Household cost allocation, enforceable conditions, compliance and realized net public benefit.

These are proposed research designs, not results. Peak-water capacity and hourly marginal-emissions preprints motivate the first two questions. The accepted health manuscript, local community study and German survey experiment motivate distinct exposure and governance questions; their differing designs cannot be combined into a single causal conclusion. Full annotations, version status and direct links appear in the dashboard’s Policy research view and downloadable reading list. [64] [65] [14] [12] [63]

10. Environmental assessment framework for data centers and AI services

A complete metric framework is not the same as a completed facility assessment. The following twenty indicators specify outcomes, units, boundaries, methods, evidence and limitations. Actual impacts cannot be fully quantified where meter records, supplier inventories, local exposure data or a defensible counterfactual are missing. Missing information remains unknown; it is never assigned a zero or used to certify an environmentally neutral system.

Define the functional unit before comparison: an accepted service outcome at a specified quality, latency, accessibility and error-risk level. Report both annual absolute totals and intensity per useful outcome. Include training, tuning, inference, retrieval, storage, retries, human correction, networks and devices when material, together with equipment production, facility construction, maintenance and end-of-life. State exclusions and allocation rules explicitly. A California purchaser may rely on facilities in other states; procurement location cannot substitute for hosting geography. [4] [6] [17] [19]

Assessment has two linked views. Attributional inventories allocate the footprint of existing activities; consequential studies estimate what changes relative to a counterfactual. Annual electricity inventories, market-based contractual claims and hourly marginal grid impacts must remain distinct. Avoid counting both supplier totals and their allocated customer share in the same system total. Assess energy-water tradeoffs, drought and heat-wave coincidence, local pollution, equipment replacement and demand rebound before accepting an efficiency claim. [60] [59] [3] [9] [65]

Measurement maturity and interpretation

Established operational measurement. Electricity, flow, fuel and basic noise indicators have established instruments. The main obstacles here are public access, matching boundaries and time resolution.

Established methods, allocation required. GHG inventories, PUE/WUE and life-cycle accounting are tractable, but comparability depends on definitions, denominator matching, supplier data and uncertainty analysis.

Local assessment required. Exposure, effluent impacts, hydrology and biodiversity require source-specific field data and models. State averages or general literature cannot substitute for these inputs.

Counterfactual and distributional analysis. Marginal emissions, net public benefits and AI rebound depend on alternatives and affected populations. Report assumptions and separate indicators rather than a universal score.

Absolute electricity use - MWh/site/year; kWh/workload

Purpose and boundary. Absolute demand determines infrastructure scale; an efficient facility can increase electricity use when workloads and installed capacity grow. Operational electricity for the named campus and reporting period; distinguish IT, cooling, power losses and other support loads. [25] [39] [1] [66]

Measurement. Sum calibrated import and on-site generation meters, reconcile exports and storage, and avoid counting generator electricity twice. **Suggested cadence:** Retain hourly readings where available, reconcile monthly bills, publish annual totals, and report missing intervals and estimated shares. **Minimum evidence:** Campus boundary, meter locations, reporting dates, grid imports, generator output, storage treatment, estimation share and utility reconciliation records.

Decision use. Compare expansion plans with observed demand trajectories and separately assess which operational changes reduce total electricity use over time. **Tradeoff or limitation:** Reducing cooling energy may increase water consumption; distinguish genuine demand reduction from shifting activity to another facility or supplier. **Current coverage:** EPRI supplies modeled state electricity, not site meters; literature benchmarks

provide selected workload measurements, with no complete campus inventory.

Peak load, timing and flexibility - MW; hourly MWh; verified curtailed MWh

Purpose and boundary. Grid capacity and reliability depend on when loads occur, while annual electricity totals can hide concentrated local infrastructure pressure. Facility demand at the grid connection and relevant utility or balancing area, distinguishing contracted, installed and actual coincident peaks. [25] [26] [43] [65]

Measurement. Use interval meters to identify seasonal peaks; compare dispatch events with documented baselines and report rebound demand after curtailment. **Suggested cadence:** Archive hourly or finer data, review monthly peaks, and assess summer, winter and grid-stress events separately from annual consumption. **Minimum evidence:** Interval load series, timezone, interconnection boundary, seasonal system peaks, baseline method, dispatched reductions, generator operation and subsequent recovery load.

Decision use. Evaluate interconnection sizing, demand response and workload scheduling with utility-specific constraints and tested performance rather than promised flexible capacity. **Tradeoff or limitation:** Shifting computation may reduce peak demand but move water use, latency or emissions to a different hour or location. **Current coverage:** EPRI includes modeled peak GW and scenarios; this dataset lacks metered hourly loads, coincident utility peaks and demonstrated flexibility.

Facility overhead efficiency: PUE - Dimensionless total facility energy / IT energy

Purpose and boundary. PUE isolates support overhead, enabling cooling and electrical-loss diagnosis while leaving workload usefulness and total demand to separate metrics. Facility and IT energy measured over identical periods and declared measurement points; include all relevant energy inputs consistently. [66] [39] [1]

Measurement. Divide facility energy by IT energy, disclose metering category and weather/utilization context, and retain the absolute numerator and denominator. **Suggested cadence:** Report annual energy-weighted PUE alongside monthly values; investigate seasonal changes and alterations in occupancy, IT loading or cooling operation. **Minimum evidence:** Facility and IT energy meters, synchronized periods, metering category, allocation for shared buildings, occupancy, weather and documented calculation method.

Decision use. Use PUE to compare operational improvements within compatible boundaries, paired with absolute electricity, water and useful-service output measures. **Tradeoff or limitation:** Improved PUE can accompany greater evaporation or more wasteful IT activity; it does not establish lower total environmental burden. **Current coverage:** The dataset explains PUE but lacks matched facility and IT meter series for Indiana or California project-level comparisons.

Purchased-electricity GHG inventory - tCO2e/year; gCO2e/kWh, separately by method

Purpose and boundary. Two accounting views describe physical grid context and contractual procurement; annual renewable claims alone cannot demonstrate carbon-free electricity every hour. Purchased electricity within the reported campus or organization; keep location-based and market-based inventories separate and disclose geographic matching. [60] [61] [59]

Measurement. Multiply electricity by documented grid-average factors for location accounting; apply qualifying contractual instruments separately, preserving residual-mix and factor provenance. **Suggested cadence:** Publish annual inventories with factor vintage and contractual coverage; add hourly matching diagnostics when time-resolved energy and supply exist. **Minimum evidence:** Electricity totals, factor geography and year, gases and GWP version, instrument ownership and retirement, coverage gaps and accounting rules.

Decision use. Assess inventory disclosure and procurement quality without subtracting claimed avoided emissions or treating certificates as proof of local zero emissions. **Tradeoff or limitation:** Contractual reductions may differ

from local hourly emissions changes; renewable procurement also has land, materials and water consequences elsewhere. **Current coverage:** Existing GHG guidance supports methods; no matched campus load, factors and instruments are available here to calculate audited site inventories.

Consequential electricity emissions - Change in tCO2e versus explicit counterfactual

Purpose and boundary. A decision can change emissions differently from its allocated inventory, especially when new capacity or different generating units respond. Power-system response to an identified load, siting, scheduling or procurement decision, over a stated geography and time horizon. [65] [61] [9]

Measurement. Model incremental generation and capacity changes against an explicit baseline; label marginal factors, dispatch assumptions and sensitivity ranges. **Suggested cadence:** Assess each material decision; update scenarios when grid forecasts, load shape or transmission assumptions change, retaining previous versions for comparison. **Minimum evidence:** Counterfactual, decision horizon, hourly loads, marginal-emissions model, capacity response, transmission constraints, model vintage and transparent sensitivity analysis.

Decision use. Compare alternatives with common service requirements and uncertainty ranges; report consequential results separately from location-based and market-based inventories. **Tradeoff or limitation:** Short-run dispatch savings may conflict with long-run infrastructure effects, local air pollution or water demand; scenario choices influence conclusions. **Current coverage:** Marginal-emissions preprint offers methods; existing state scenarios do not identify causal grid responses or establish site-specific avoided emissions.

Embodied and full life-cycle GHG - tCO2e/project; kgCO2e/defined service

Purpose and boundary. Operational decarbonization leaves equipment and construction burdens; omitting manufacturing can substantially understate the footprint of a specific AI deployment. Raw materials, fabrication, buildings, equipment, transport, operation and end-of-life; identify networks and user devices included or excluded explicitly. [4] [19] [67] [10]

Measurement. Combine process inventories with operational data; allocate shared hardware transparently across service life and utilization, testing alternative lifetimes and allocations. **Suggested cadence:** Update at procurement, retrofit and retirement, then reconcile annually with asset records and significant changes in utilization or supplier data. **Minimum evidence:** Bill of materials, supplier inventories, asset lifetimes, utilization, functional unit, allocation rules, electricity assumptions, end-of-life treatment and uncertainty ranges.

Decision use. Compare replacement, refurbishment and deployment options using equivalent service requirements, consistent system boundaries and separate life-cycle stage results. **Tradeoff or limitation:** New hardware can reduce operational energy yet increase manufacturing impacts; lifetime extension benefits depend on performance, reliability and grid emissions. **Current coverage:** BLOOM and computing studies illustrate allocated life-cycle estimates; project bills of materials and comparable supplier inventories are absent here.

Direct fuel and refrigerant emissions - Fuel quantity; kg refrigerant lost; tCO2e/year

Purpose and boundary. Backup generation and cooling-fluid leaks remain direct emissions sources even when annual purchased electricity is matched with renewable instruments. On-site combustion and refrigerant releases from campus equipment, including maintenance and retirement; separate direct emissions from purchased electricity. [66] [68] [28]

Measurement. Calculate combustion from fuel records and factors; estimate refrigerant losses through charge, additions, recovery and inventory reconciliation by substance. **Suggested cadence:** Record every refueling, refrigerant service and leak event; reconcile annual stocks and report testing, outage and routine operating hours. **Minimum evidence:** Fuel invoices, engine runtime, refrigerant species and charge, service additions, recovered

mass, emission factors, GWP version and boundary reconciliation.

Decision use. Prioritize leak prevention and low-emission backup options; assess actual use separately from installed engine capacity and maximum permitted operation. **Tradeoff or limitation:** Air-cooled or mechanical cooling may reduce water demand while changing electricity use and refrigerant exposure; backup choices change reliability and pollution. **Current coverage:** Existing sources describe designs and reporting concepts; no complete campus fuel ledger, runtime history or refrigerant mass balance is included.

Direct withdrawal by source - m^3/year; m^3/day; source-specific volumes

Purpose and boundary. Withdrawal indicates pressure on supply and infrastructure, but does not reveal how much returns or whether the source is locally scarce. Water entering the named campus from surface water, groundwater, municipal or reclaimed supplies; report construction dewatering separately from operations. [2] [28] [42] [45] [56]

Measurement. Meter each intake, identify ultimate source and treatment, and distinguish withdrawals from storage capacity, sewer flows and water consumed. **Suggested cadence:** Publish monthly totals, daily peaks and seasonal source mix; retain shorter intervals during drought, construction pumping or supply restrictions. **Minimum evidence:** Intake meters, source identities, daily and monthly series, construction versus operation separation, estimation flags and utility delivery reconciliations.

Decision use. Assess water-supply agreements and drought performance using actual demand by source, independently from treatment plant capacity and permit design maxima. **Tradeoff or limitation:** Reclaimed water can reduce potable demand while altering treatment energy and downstream return flows; source substitution requires a watershed-specific assessment. **Current coverage:** Google discloses selected 2024 campus volumes, sometimes estimated; the sample contains no Indiana or California campus water measurements.

Direct consumption and stated WUE - m^3/year; L/kWh IT, numerator explicitly labeled

Purpose and boundary. Consumptive losses reduce water available to other users; WUE supports operational diagnosis only when its numerator and denominator are consistent. Water not returned within the stated basin and time boundary; distinguish site water-input WUE from consumption-based intensity measures. [2] [28] [66]

Measurement. Reconcile withdrawals, discharge destinations and storage changes; divide the explicitly named water numerator by matched IT electricity for intensity. **Suggested cadence:** Reconcile monthly water balances and publish annual WUE with summer peaks; separately report uncertain returns, storage changes and measurement gaps. **Minimum evidence:** Withdrawal, discharge and storage records, basin and timing of returns, IT energy, balance closure, water quality and explicitly stated WUE definition.

Decision use. Compare cooling alternatives using both absolute volumes and defined intensities; do not calculate WUE from state electricity and unrelated campuses. **Tradeoff or limitation:** Low site WUE can shift demand to electricity generation; low consumption does not imply harmless discharge temperature or chemistry. **Current coverage:** Google provides withdrawal-minus-discharge estimates for named sites; missing matched IT electricity prevents comparable campus WUE calculations in this dataset.

Indirect electricity and supply-chain water - m^3 withdrawal and m^3 consumption, separately

Purpose and boundary. Cooling alone misses water used by power plants and semiconductor production, potentially shifting burdens to distant communities and ecosystems. Electricity-generation and hardware-manufacturing water attributed to the service; retain upstream watersheds and distinguish these burdens from direct site water. [3] [5] [15] [67]

Measurement. Multiply location-specific energy or material activity by compatible withdrawal and consumption factors; disclose cooling, hydropower and allocation assumptions. **Suggested cadence:** Recalculate annually and when electricity sourcing, hardware procurement or operating location changes; use seasonal factors where validated data permit. **Minimum evidence:** Energy geography, generation mix, water-factor definitions and vintage, supplier process data, basin assignments, allocation method and uncertainty ranges.

Decision use. Evaluate cooling and procurement together, reporting direct and indirect components and testing alternative electricity-water factors before selecting locations. **Tradeoff or limitation:** Water-intensive clean electricity or fabrication can offset site savings; upstream allocation choices can dominate cross-location comparisons and apparent rankings. **Current coverage:** Studies offer modeled indirect-water examples; supplier inventories and matched campus electricity are missing, preventing reliable project-specific totals here.

Seasonal scarcity and supply resilience - Monthly m^3; basin stress category; drought scenarios

Purpose and boundary. Equal annual volumes can create different harms depending on dry-season demand, groundwater recovery, existing users and ecological flow requirements. Facility and upstream watersheds, local users and ecological flows during relevant months; distinguish water availability from treatment and pipe capacity. [62] [3] [43] [47]

Measurement. Overlay source-specific seasonal demand with basin screening indicators, validated local hydrology and drought scenarios; avoid interpreting screening scores as spare supply. **Suggested cadence:** Review seasonal conditions monthly, update drought assessments as conditions change, and reassess before expansions or changes in water source. **Minimum evidence:** Source watershed, monthly withdrawals and consumption, competing demand, groundwater or streamflow data, drought assumptions, ecological requirements and local hydrologic review.

Decision use. Screen sites, then request local drought-yield analysis, operating triggers and contingency plans before concluding that infrastructure can support expansion. **Tradeoff or limitation:** Lower-carbon locations may face greater water scarcity; annual replenishment claims may occur in different basins or seasons from consumption. **Current coverage:** Aqueduct supplies screening context; Indiana system capacities do not establish sustainable yield, and this dataset lacks calibrated campus water models.

Combustion pollution and population exposure - kg NOx/PM2.5; ug/m^3; separately modeled health outcomes

Purpose and boundary. Pollution burdens depend on where emissions travel and who is exposed; equal emissions totals can produce unequal potential health impacts. On-site engines plus attributed power-plant pollution and exposed receptors; separate emitted mass, modeled concentration, observed exposure and health outcomes. [68] [13] [14] [22] [23]

Measurement. Combine runtime and tested or factor-based emissions with dispersion and population data; validate concentrations and retain uncertainty in health models. **Suggested cadence:** Log engine events continuously, summarize emissions annually, and sample relevant background, testing and outage periods with meteorology and receptor locations. **Minimum evidence:** Engine specifications, fuel and hours, stack tests or factor quality, meteorology, background concentrations, receptor populations and documented exposure-response assumptions.

Decision use. Compare backup and siting options through exposure analysis; prioritize monitoring near sensitive receptors and avoid attributing individual illness without evidence. **Tradeoff or limitation:** Carbon-oriented scheduling or backup substitution can move local pollutant exposure; modeled health damages are not observed

diagnoses or proof of causation. **Current coverage:** Health studies provide modeled burdens and research gaps; this dataset contains no causal epidemiology or campus-specific verified exposure series.

Residential noise and tonal character - LAeq/Lmax dBA; LCeq dBC; spectrum and tonality

Purpose and boundary. Persistent tones and low-frequency sound can matter beyond a single average decibel value; facility-internal measurements do not describe residential exposure. Outdoor and relevant indoor residential receptors exposed to cooling, generators and construction; specify background conditions, time periods and equipment operation. [69] [27] [13] [23]

Measurement. Measure calibrated A- and C-weighted levels, frequency spectra and tonal characteristics; compare baseline and operating periods under documented meteorology. **Suggested cadence:** Collect day and night baselines, commissioning measurements and seasonal operating surveys; investigate complaints and intermittent generator events with time-stamped records. **Minimum evidence:** Receptor map, calibrated instruments, LAeq and maximum periods, LCeq, frequency bands, background subtraction, meteorology, runtime and applicable local criteria.

Decision use. Assess design mitigation and actual neighbor experience against verified local criteria; do not import occupational limits as residential standards. **Tradeoff or limitation:** Quieter cooling or barriers can affect energy use, airflow and cost; acoustically equal levels can have different perceived characteristics. **Current coverage:** Community studies identify concerns; no standardized Indiana or California receptor measurements are included, so compliance or health effects cannot be scored.

Effluent heat and chemistry - m^3/day; °C/temperature rise; pH; mg/L and kg/day

Purpose and boundary. Returning water does not eliminate impacts if temperature, salts, biocides or other constituents affect treatment systems or aquatic ecosystems. Campus wastewater, cooling blowdown and receiving treatment or water bodies; distinguish sewer design capacity from measured outfall quality and quantity. [70] [56] [2]

Measurement. Pair flow with temperature and chemical sampling to calculate loads; evaluate receiving-water conditions, treatment performance and source-specific chemicals used. **Suggested cadence:** Follow applicable permit schedules; proposed assessment adds representative seasonal and upset-event samples, with continuous flow and temperature where feasible. **Minimum evidence:** Actual flow, sampling location and method, temperature, chemistry, chemical inventory, receiving conditions, treatment destination, laboratory quality controls and applicable permit.

Decision use. Evaluate cooling and wastewater designs using measured loads and receiving conditions; obtain the specific permit before assessing compliance or limits. **Tradeoff or limitation:** Water reuse can reduce withdrawal while concentrating constituents and increasing treatment demand; disposal pathways determine where remaining burdens occur. **Current coverage:** Indiana sewer approval supplies design flows only; no campus discharge temperature, chemistry or receiving-water monitoring is available in this dataset.

Land change and biodiversity - Hectares disturbed; habitat condition and connectivity indicators

Purpose and boundary. A small campus footprint can depend on extensive infrastructure or affect connected habitats; hectares alone omit ecological condition and sensitivity. Campus, substations, transmission, generation and relevant supply-chain land; distinguish previously developed land, sensitive habitats and indirect infrastructure changes. [71] [67] [9] [22]

Measurement. Compare mapped pre-project land cover with surveyed habitat and post-project conditions; report fragmentation and species-specific findings without collapsing ecological differences. **Suggested cadence:** Establish a preconstruction baseline, monitor construction and restoration, then survey appropriate ecological seasons after operation and major site changes. **Minimum evidence:** Project and associated infrastructure maps, baseline land cover, seasonal habitat surveys, connectivity assessment, mitigation commitments and independently checked restoration outcomes.

Decision use. Compare siting alternatives and avoidance measures using habitat evidence; keep biodiversity conclusions distinct from greenhouse-gas or fiscal performance. **Tradeoff or limitation:** Renewable generation and grid upgrades can reduce operational carbon while expanding land demand; restoration pledges require location-specific outcome verification. **Current coverage:** The present dataset lacks parcel-level ecological baselines or infrastructure footprints; this domain remains an unfilled project-assessment requirement, not zero impact.

Hardware turnover, e-waste and critical materials - Tonnes/year by route; lifetime years; material-specific kg

Purpose and boundary. Rapid hardware replacement increases manufacturing demand and waste; critical-material recovery and safe handling depend on traceable downstream processes. Servers, accelerators, storage, network equipment and batteries from procurement through reuse, recycling and final disposal, including material supply chains. [10] [19] [67] [22]

Measurement. Reconcile asset and mass inventories; track service life, reuse destination and recovery by material, distinguishing collection claims from verified recycling outcomes. **Suggested cadence:** Record procurement and retirement events, audit downstream transfers annually, and update material inventories when equipment configurations or suppliers change. **Minimum evidence:** Asset register, equipment mass and composition, actual lifetimes, refurbishment records, waste manifests, downstream destinations and evidence of material recovery.

Decision use. Choose repair, refurbishment and procurement options using life-cycle comparisons; require downstream documentation and material-specific reporting for retired assets. **Tradeoff or limitation:** Extending hardware life can reduce manufacturing demand while increasing operational energy; recycling rates alone can conceal losses of specific critical materials. **Current coverage:** Generative-AI research supplies global waste scenarios; no local asset mass balance or verified supplier and recycler chain is included.

Cumulative burden and distributional equity - Separate exposure indicators by population and geography

Purpose and boundary. A project-wide average can hide disproportionate burdens on particular neighborhoods or workers, while benefits may accrue to different populations. Nearby residents, workers and upstream communities, including sensitive populations; compare distribution of burdens and benefits within a defined assessment geography. [82] [12] [22] [23]

Measurement. Overlay measured or modeled exposures with demographic vulnerability and existing burdens; involve residents and preserve each indicator rather than inventing a composite. **Suggested cadence:** Assess before siting, update annually with disclosed data vintages, and review when exposures, population conditions or expansion plans materially change. **Minimum evidence:** Spatial exposure estimates, background burden, population and worker characteristics, data vintages, community input, geographic uncertainty and separately reported benefits.

Decision use. Identify priority monitoring and mitigation areas through source-specific analysis and community participation; avoid transferring California percentile ranks to Indiana. **Tradeoff or limitation:** Low statewide carbon can coexist with local pollution or water hardship; screening percentiles do not prove project-attributable illness or individual exposure. **Current coverage:** CalEnviroScreen 5.0 supports California screening, but no campus-to-tract overlay or community exposure inventory has been completed in this dataset.

Public costs, jobs and fiscal distribution - Inflation-stated dollars; job-years; permanent FTE; rate effects

Purpose and boundary. Large announced investment does not establish net community benefit; infrastructure costs and financial risks can shift among customers and taxpayers. Public budgets, utility customers, workers and host community over the project life; distinguish private investment from net local public benefit. [48] [49] [36] [34] [38]

Measurement. Compare verified taxes and payroll with incentives, infrastructure, services and stranded-cost scenarios; separate construction job-years from ongoing full-time employment. **Suggested cadence:** Assess before agreements, verify annual realized jobs and revenue, and stress-test delay, closure or lower-than-promised demand throughout contract periods. **Minimum evidence:** Executed agreements, incentive schedules, verified payroll and employment, actual tax receipts, upgrade costs, financing terms and utility cost-allocation evidence.

Decision use. Evaluate benefit agreements and utility cost allocation using common fiscal assumptions, distribution tables and documented responsibilities if projects underperform. **Tradeoff or limitation:** Jobs and revenue may coexist with environmental burdens; monetizing selected harms cannot replace separate health, water or biodiversity assessment. **Current coverage:** Indiana announcements provide job and tax commitments; tariff examples show allocation mechanisms, but no independent project net-benefit calculation is available.

Disclosure, verification and accountability - Coverage percentages; reporting lag; unresolved cases and resolution time

Purpose and boundary. Comparisons fail when reporting omits campuses or changes definitions; clear evidence status prevents promises and design permits becoming apparent measurements. Every reported metric, campus boundary and affected community; assess data completeness and accountability separately from environmental performance itself. [17] [12] [28] [38]

Measurement. Track missing data, measurement versus estimation, assurance scope, public access and complaint outcomes; preserve versioned sources and correction histories. **Suggested cadence:** Update a public ledger quarterly where feasible, reconcile annually, and disclose material boundary changes, incidents and corrections promptly. **Minimum evidence:** Metric definitions, coverage denominator, data owner, reporting dates, raw-to-reported reconciliation, estimation flags, independent review scope and complaint-response records.

Decision use. Set procurement and community reporting expectations, identify evidence gaps, and assign responsible owners without treating disclosure quality as low environmental impact. **Tradeoff or limitation:** More reporting improves scrutiny but has costs and confidentiality constraints; redaction should preserve decision-relevant aggregates and transparent explanations for gaps. **Current coverage:** This review preserves source status and uncertainty; project-level reporting coverage, regular complaint logs and independent assurance remain incomplete.

Useful AI service and rebound - kWh/accepted task; service quality; total annual footprint

Purpose and boundary. Lower energy per request can coincide with higher total demand; comparing unequal tasks or unconstrained output lengths can mislead procurement. Equivalent end-user service across training, tuning, inference, idle capacity and associated equipment; include deployment volume and any claimed downstream benefit. [6] [4] [8] [9] [17]

Measurement. Measure energy for fixed quality and latency requirements, then multiply by actual volume; test counterfactual savings and induced additional demand. **Suggested cadence:** Benchmark before selection, remeasure after model or hardware changes, and review monthly service volume and annual operational and embodied totals. **Minimum evidence:** Representative task set, acceptance criteria, model and hardware versions, tokens, batch size, latency, energy boundary, utilization and annual workload volume.

Decision use. Procure the least-burden option meeting service requirements, monitor absolute budgets, and report avoided impacts separately with a defensible baseline. **Tradeoff or limitation:** Smaller models, caching and batching may change quality, privacy or latency; efficiency gains may stimulate demand that erodes aggregate savings. **Current coverage:** Published benchmarks and BLOOM illustrate bounded cases; this dataset cannot rank current vendors or calculate a universal footprint per prompt.

The dictionary is a synthesis of referenced methods and analyst recommendations. Monitoring frequencies are suggested practice unless a cited jurisdiction-specific instrument imposes a schedule. Significance thresholds require site context, an applicable permit or law, and explicit policy judgment. The twenty indicators are not weighted into one environmental score: lower carbon cannot automatically compensate for a local water shortage, pollution exposure, habitat loss or inequitable cost shift.

11. Application to California policy and the CivicShield concept

California's new data-center legislation was signed in September 2026. Enactment is separated here from ordinary effective dates, agency implementation deadlines and project triggers. The policy register below links the required or decision-relevant measurements to their authority and limitations. It is a research mapping, not a determination that a particular project complies. [86]

CEC forecasting, benchmarking and generation review

Existing programs. California electricity planning, covered buildings and onsite thermal generation. IEPR forecasts incorporate utility data-center applications. Covered buildings over 50,000 square feet report annual energy use. Thermal generation of 50 MW or more falls within CEC licensing jurisdiction. [72]

Assessment application. Connect load forecasts with procurement, local permitting and backup-generator impacts. **Boundary:** CEC does not permit data centers themselves. A small-power-plant exemption still requires environmental review and subsequent permits. **Research question:** How do projected connections compare with realized demand and local pollution?

AB 1577: facility energy reporting

Enacted; effective 2027-01-01. Data centers ≥10 MW; specified public, research and utility facilities excluded. Requires energy, peak-load, PUE and fuel reporting; CEC sets filing timing. Public releases are anonymized aggregates; IEPR load-trend assessments start in 2029. [73]

Assessment application. Local discretionary applications must estimate electricity, onsite generation and boundary noise. **Boundary:** Facility reporting does not isolate AI workloads; customer energy data remain protected. **Research question:** Can aggregate disclosures reveal infrastructure pressure while retaining useful local accountability?

AB 2469: water conditions for development

Enacted; effective 2027-01-01. Construction or expansion increasing maximum peak water use; discretionary and ministerial approvals. Requires water-supply assessment, projected use, efficiency and workforce disclosures, and applicant funding of necessary water infrastructure. Water-scarcity plans begin January 1, 2028. [75]

Assessment application. Evaluate dry-year reliability alongside project demand and infrastructure costs. **Boundary:** No MW cutoff appears here; satisfying conditions does not require approval. **Research question:** Are peak cooling demands compatible with enforceable drought reductions and existing customers' needs?

AB 2619: water-use disclosure

Enacted; effective 2027-01-01. Data centers across defined size classes, including facilities below 2 MW. Initial licensing requires good-faith water estimates and source disclosure; renewals report prior-year water use, direct use by source and cooling type. [76]

Assessment application. Supplier shortage planning must include unconstrained demand from data centers and other large users. **Boundary:** Withdrawals or deliveries are not automatically consumptive losses or upstream electricity water use. **Research question:** How accurately do license estimates predict metered seasonal demand?

SB 886: interconnection and cost protection

Enacted; effective 2027-01-01. Participating data-center customers; transmission-specific provisions cover new agreements from January 1, 2027. Requires tariffs or updated rules by January 1, 2028 to prevent stranded costs and cost shifts. Transmission provisions allocate upgrade costs, constrain refunds and require termination protection. [77] [74]

Assessment application. Compare committed demand, infrastructure exposure and demand-response participation. **Boundary:** Implementation requires CPUC decisions; transmission cost allocation is constrained by federal law. **Research question:** Do deposits and exit obligations cover realistic cancellation and underutilization scenarios?

AB 2383: service tariffs and procurement

Enacted; effective 2027-01-01. Participating customers; generation-tariff thresholds must be set no higher than 25 MW. Requires cost-protective service tariffs, clean-energy-consistent procurement and generation payment mechanisms lasting at least ten years. CCA/provider generation tariffs are due January 1, 2028. [74]

Assessment application. Link added load with incremental procurement, collateral and societal-cost contributions. **Boundary:** Generation-specific exemptions and thresholds are not universal exemptions from every provision. **Research question:** Which contract terms best prevent stranded procurement without weakening clean-energy requirements?

SB 1168: rate-structure assessment

Enacted; effective 2027-01-01. Data-center transmission and distribution connections in CPUC proceedings. Requires assessing opportunities for rates that assign reasonable network costs and proportionate load-growth procurement costs to data centers while relieving residential rate pressures. [79]

Assessment application. Connect resource planning to distributional effects on households, including CARE and FERA participants. **Boundary:** The act directs assessment; it neither publishes a completed tariff nor establishes a numerical cost allocation. **Research question:** How do allocation choices affect low-income customers under different load-growth outcomes?

SB 887: CEQA review and conditional streamlining

Enacted; effective 2027-01-01. Data-center entitlements; categorical-exemption restrictions contain specified facility exceptions. Extends CEQA coverage and bars categorical exemptions for covered projects. Leadership-project streamlining requires demanding energy, water, cost-protection and community-benefit conditions. [78]

Assessment application. Streamlining conditions include hourly zero-carbon supply within five years, predominantly new resources, and efficient cooling. **Boundary:** These streamlining conditions are not universal operating standards for every data center. **Research question:** Do enforceable project conditions reduce cumulative burdens beyond statutory minimum review?

SB 57: earlier cost-shift assessment

Enacted; effective 2026-01-01. Electrical-corporation costs caused by new data-center loads. Authorizes assessment of procurement, network investment and stranded-asset cost shifts. Any completed assessment must be submitted to legislative committees and posted by January 1, 2027. [80]

Assessment application. Provides an analytical bridge between electricity growth, infrastructure risk and ratepayer protection. **Boundary:** SB 57 authorizes assessment; it does not itself order a new tariff. Section sunsets January 1, 2031. **Research question:** Which documented cost risks warrant specific safeguards in later tariff proceedings?

AB 93: vetoed water proposal

Vetoed; no effective date. Historical data-center water-policy proposal. The Governor returned AB 93 without approval. It must not be displayed as an operative water-reporting mandate. [81] [75] [76]

Assessment application. Compare the unsuccessful 2025 proposal with the separately enacted 2026 water-disclosure laws, AB 2469 and AB 2619. **Boundary:** A vetoed proposal supplies legislative context, not compliance authority; later bills require independent scope and effective-date checks. **Research question:** What changed between the vetoed proposal and the enacted water protections?

CalEnviroScreen 5.0: cumulative-burden screening

Final tool; not a facility permit. California census-tract pollution burden and population vulnerability. Updates screening data and methods, adding diabetes prevalence and small air-toxic sites. It informs CalEPA's final September 18, 2026 disadvantaged-community designation. [82]

Assessment application. Screen proposed sites, then investigate nearby receptors, emissions, noise and community priorities. **Boundary:** A screening percentile neither measures a project's incremental health risk nor determines permit approval. **Research question:** Would the project add burdens where exposure and vulnerability already overlap?

DWR: supplier shortage and drought planning

Existing annual reporting. Urban suppliers exceeding 3,000 connections or 3,000 acre-feet annually. Requires annual supply-demand assessment and shortage reporting, generally by July 1; specified imported-water allocation timing can extend submission. [83]

Assessment application. Use supplier assessments, shortage-contingency plans and drought-risk assessments to contextualize new cooling demand. **Boundary:** Regional or statewide sufficiency does not establish site-specific supply. Assessments require ongoing updates as actual conditions change. **Research question:** What restrictions apply when facility demand peaks during a supplier shortage?

SB 253: corporate emissions disclosure

Enacted statute; subsequent regulatory approval unverified. U.S. entities doing business in California with annual revenue above $1 billion. Requires Scope 1-2 reporting in 2026 and Scope 3 from 2027. CARB's September notice identifies November 10, 2026, conditional on OAL approval. [84]

Assessment application. Use corporate emissions to supplement facility and supply-chain evidence. **Boundary:** Corporate totals are not facility or AI-model measurements. This review did not establish whether OAL approved the regulation after CARB’s September 1 notice. **Research question:** Can corporate inventories be reconciled with location-specific electricity and equipment emissions?

California’s EPRI historical model estimate is 9.058 TWh in 2024, with source-variation bounds of 8.758-9.357 TWh and modeled peak demand of 1.069 GW. The 2030 low, medium and high scenarios are 10.793, 15.027 and 19.824 TWh. These are data-center-plus-cryptocurrency estimates, not measured AI loads or confidence intervals. No California campus appears in the harmonized Google water sample, so it cannot supply a California water total or WUE. A state electricity number also cannot determine local groundwater, air quality, noise or health outcomes.

[25] [26] [28]

The Indiana application reaches a parallel conclusion about incompatible boundaries: a campus annual water projection, a sewer design peak, municipal plant capacity and construction dewatering do not form a measured consumption series. The correct next step is a shared facility and utility inventory with monthly source-resolved balances, hourly hot-weather peaks, baseline wells, discharge and construction pumping kept distinct. The dashboard's application view records the values supported by current evidence and the additional data needed. [45] [56] [43] [46]

The supplied June 2026 CivicShield WIA and NAWEI artifacts adapt environmental review into workforce governance. Their useful components are purpose and need, alternatives, affected tasks, consultation, assigned mitigation and monitoring. They contain no measured physical footprint. Their unvalidated score weights differ between the plan and prototype, and the prototype's multiplication of composite risk by worker count does not establish exposed or displaced workers. These are findings from the supplied design artifacts, not claims about a published validated instrument.

CivicShield crosswalk: Purpose, baseline, alternatives

Use equal service volume and quality across alternatives. Compare baseline versus deployment changes in task-hours, employment, earnings, service outcomes, energy, emissions, and water. **Interpretive limit:** The baseline and counterfactual must be documented; current staffing is not automatically the future without AI.

CivicShield crosswalk: Exposure and consequence

Separate hazard, exposure, severity, likelihood, duration, and reversibility. Record affected task-hours and workers independently from displacement, wage loss, safety, and service failures. **Interpretive limit:** Exposure is not displacement; a composite score is not a probability.

CivicShield crosswalk: Tiered review

Add documented significance criteria and escalation triggers for concentrated, irreversible, cumulative, or uncertain harms. Permit human escalation regardless of aggregate score. **Interpretive limit:** Bands are proposed governance choices, not validated thresholds or legal clearance.

CivicShield crosswalk: Distribution and cumulative burden

Evaluate repeated deployments across employers and regions, overlapping transition burdens, affected communities, and unequal access to training or clean infrastructure. **Interpretive limit:** Protect privacy and report subgroup uncertainty; regional correlation does not prove individual harm.

CivicShield crosswalk: Mitigation and residual effects

Compare pre-control and residual impacts; track implementation, worker retention, real earnings, placement quality, and environmental reductions against explicit targets. **Interpretive limit:** Training participation is not verified employment recovery; modeled improvement is not achieved mitigation.

CivicShield crosswalk: Lifecycle physical footprint

Add training and inference energy, electricity emissions, direct and indirect water consumption, hardware manufacturing, replacement, and waste; identify site, time, allocation, and supplier evidence. **Interpretive limit:** Avoid double counting; report absolute totals and useful-service intensity separately.

CivicShield crosswalk: California policy and public accountability

Develop pilot procurement disclosures, alternatives memoranda, accountable mitigation commitments, public monitoring, and reassessment when scope changes. **Interpretive limit:** These are policy options. The pitch's N-6-26 reference and any legal requirements require current primary-source verification.

For procurements governed by California's State GenAI procurement policies, a completed SIMM 5305-F risk assessment is required regardless of risk level, as clarified by CDT Technology Letter 24-03. A proposed environmental and workforce annex could add equal-service alternatives, attributable vendor resource data, hosting locations, lifecycle allocation, worker outcomes, absolute budgets, accountable controls and reassessment after material changes. This annex is a recommendation; it is not an existing universal mandate, a replacement for CEQA or permits, or a claim that NAWEI scores satisfy California procurement requirements. [85]

A useful pilot compares the current workflow, process redesign, conventional software, a smaller model and the proposed system using the same service target. Collect task-hours, retention and inflation-adjusted earnings separately from electricity, lifecycle emissions and seasonal water. Measure actual follow-up outcomes rather than translating abstract risk scores into predicted headcounts. Report any external societal emissions savings against a separate counterfactual and evaluate whether increased demand uses up operational efficiency gains. [9] [17] [19]

Priority implementation research should test whether California reporting improves independent reproducibility; whether tariff and contract rules allocate realized costs as intended; whether project disclosures capture cumulative local burdens; and whether AI procurement pilots deliver useful service with lower absolute physical impacts and acceptable worker outcomes. Predefine outcomes, comparison groups, confounders, uncertainty and material changes that reopen review.

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